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Key Points:

- We perform simulations that model the growth of a mixed layer front forced by a background deformation flow
- It begins in a low Rossby number, mesoscale regime, exhibits a submesoscale transient phase before reaching a final, arrested state
- In the presence of a background strain flow, the final state is separated into a resolved arrested front and an unmixed entrainment region

Supporting Information:

Supporting Information may be found in the online version of this article.

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Formation and Arrest of a Surface Density Front via Strain-Driven Frontogenesis

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Abstract Ocean fronts play a key role in the vertical transport of heat, energy and biogeochemical tracers, but their behavior and life cycle is not well-understood, especially at smaller scales. We present a large-eddy simulation of mixed layer frontogenesis capturing the onset of a submesoscale regime from a larger scale, rotation-dominated front. Frontogenesis shifts the balance between vertical mixing and restratification driven by frontal ageostrophic secondary circulation and the final state is a combination of a small, turbulent arrested region and the restratified remainder of the front, which sits above an unmixed, stable region. The transition to this state is marked by the appearance of down-front symmetric cells, which coalesce to form a circulation that entrains fluid from the mixed layer base. We describe the results in the context of submesoscale instability and potential vorticity transport and discuss the implications for frontogenesis and frontal arrest in the ocean mixed layer.

Plain Language Summary Ocean fronts are boundaries between regions of ocean water with different density. They may be highly energetic, with strong vertical transport of heat and nutrients. The behavior of fronts which are smaller than ~20 km is not well-understood, but models of the physical processes that occur at this scale are necessary for simulations to correctly represent the global ocean. This study presents a simulation of an ocean front as it evolves and sharpens, when forced by a background, larger-scale flow. The front is in the upper mixed layer, where quantities are strongly mixed in the vertical due to cooling at the ocean surface—this competes with the tendency for the front to slump as it sharpens. The final state is a balance of these two effects, we demonstrate how it can be understood using the transport and destruction of unstable fluid and compare the outcome of the simulations to previous theoretical and analytical studies.

1. Introduction

The first few tens of meters below the surface of the ocean host processes that are crucial for the large-scale dynamics of the ocean and its role in the climate system, yet are poorly understood. For example, to know how much kinetic energy they dissipate would help close the ocean's kinetic energy budget, but reliable and global estimates are still lacking (Ferrari & Wunsch, 2009). The upper ocean also features significant vertical velocities, and therefore are key to understanding air-sea exchanges of heat and carbon (Aravind et al., 2023; Freilich & Mahadevan, 2021; Pham et al., 2024) or how nutrients get brought up to the sun-exposed surface (Mahadevan, 2016). Many of these processes occur in submesoscale flows, that is, flows that are 0.1–10 km wide at mid-latitudes, large enough to be constrained by rotation and stratification, but small enough to be prone to instabilities that may effect a forward cascade of kinetic energy (Brüggemann & Eden, 2015; Srinivasan et al., 2023). Submesoscale currents have recently drawn interest as their role in the global ocean has become more clear (Gula et al., 2021; McWilliams, 2016; Taylor & Thompson, 2022).

Submesoscale-resolving images of surface ocean properties show complex patterns of fronts, highly anisotropic features with large horizontal gradients of tracers, velocities and density (McWilliams, 2021). Ocean fronts and filaments—pairs of opposing fronts—are associated with certain larger scale features, for example, the edges of mesoscale eddies or separating boundary currents (McWilliams, 2021). A classical model of frontogenesis, introduced in an atmospheric context by Hoskins and Bretherton (1972), places fronts in an ambient deformation flow that is oriented such that it sharpens and extends the front. In a rotating front, an ageostrophic secondary circulation emerges that accelerates the growth of horizontal gradients at the surface which, in many reduced models, reach infinity in finite time (Barkan et al., 2019; Blumen, 2000; Shakespeare, 2016). In the real ocean, it is expected that some breakdown of the dynamics presented above will eventually provide the effective diffusion that arrests frontogenesis. Possibilities include shear-instability driven turbulence (Samelson &

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Skyllingstad, 2016) or shear dispersion due to a mixing-driven secondary circulation (Crowe & Taylor, 2019), among others, though the relative importance of each mechanism in any given situation remains an open question (McWilliams, 2021). Additionally, baroclinic and barotropic instabilities may break up the front and cause it to spin out into eddies before reaching a small across-front scale.

Layers of fluid in contact with the ocean surface are typically well-mixed vertically due to boundary-layer turbulence generated by interactions with the lower atmosphere via cooling or winds (Gula et al., 2021; Taylor & Thompson, 2022). This introduces vertical mixing into the leading-order momentum balance to produce the turbulent thermal wind (TTW) balance between vertical mixing, horizontal pressure gradient and Coriolis force. Frontal evolution in this regime is not well-understood; and it is not yet clear whether the TTW secondary circulation at a front promotes (Gula et al., 2014; Sullivan & McWilliams, 2024) or inhibits (Crowe & Taylor, 2019) frontogenesis, and it likely introduces competing effects (Dauhajre et al., 2025).

Surface forcing by cooling or winds may drive instability in a near-surface ocean flow (Brannigan, 2016; Thomas, 2005). Symmetric instability (SI) and low-frequency inertia gravity waves (IGWs) catalyze extraction of energy from the background flow to the ageostrophic secondary circulation in fronts forced by a background strain (Thomas, 2012). This suggests a pathway of extraction of energy from mesoscale eddies to submesoscales facilitated by driven SI at growing fronts. In addition, vigorous instability can produce unbalanced secondary circulations at fronts, which may act to briefly strengthen horizontal gradients (Atkinson et al., 2025; Wienkers et al., 2021).

In this paper, we present what is, to our knowledge, the first simulation of controlled deformation-driven frontogenesis featuring a resolved arrest mechanism. We take a submesoscale front in a turbulent mixed layer, driven by surface buoyancy loss, and apply a forcing by a balanced, background deformation flow to bring the front to a smaller lateral length scale. During its evolution, a pattern of thin, symmetric (i.e., invariant in the down-front direction) cells briefly emerge in the front, which act to restratify most of the mixed layer. The front is arrested at a finite width, well above the grid scale. The arrest state is characterized by a region of vigorous mixing at the surface front, with stable fluid resupplied by entrainment from the base of the mixed layer. Below this region, vertical mixing is not able to overcome restratification by the secondary circulation and the front evolves toward a tilted state, with little mixing occurring underneath the front. The simulation domain is narrow in the along-front direction with the aim of providing realistic convection but suppressing along-front instabilities to focus on the frontal secondary circulation when driven by mixing and a background strain flow. We show that this also simplifies the implementation of the background strain forcing in a three-dimensional domain.

Section 2 outlines the computational experiment and methods used. Section 3 shows the results, including evolution of the front and instability. Finally, we discuss the results, agreement with prior observation and theory, and possible implications for submesoscale frontogenesis in Section 4.

2. Methods

2.1. Model Setup

We perform simulations of the non-hydrostatic Boussinesq equations forced by both a background deformation flow $\mathbf{U}$ in cyclogeostrophic balance and surface buoyancy loss.

$$\frac{D_a \mathbf{u}}{Dt} + f \hat{\mathbf{z}} \times \mathbf{u} + \nabla p - b \hat{\mathbf{z}} - a u \hat{\mathbf{x}} + a v \hat{\mathbf{y}} = 0, \quad (1a)$$

$$\frac{D_a b}{Dt} = B, \quad (1b)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (1c)$$

where

$$\frac{D_a}{Dt} = \frac{\partial}{\partial t} + (\mathbf{u} - a \mathbf{x} \hat{\mathbf{x}}) \cdot \nabla \quad (2)$$

is an approximated material derivative of a fluid element and $\mathbf{U} = \alpha(-x\hat{x} + y\hat{y})$, where α is the strain rate, which we treat as a spatially constant. The expression for D_α/Dt follows from what we call the “thin-slice approximation” described in Text S1 in Supporting Information S1. As the along-front length of the simulation domain is small, the along-front velocity due to the deformation flow remains negligible throughout the domain, when compared with the velocity due to the frontal jet. This allows us to treat the problem as 2D frontogenesis with the inclusion of resolved vertical mixing induced by surface cooling.

The buoyancy field is b and B is the buoyancy forcing induced by surface cooling. The coordinate system is aligned with the initial conditions, described in the Text S2 in Supporting Information S1, such that x , y and z are the across-front, down-front and vertical directions respectively and $\nabla = (\partial_x, \partial_y, \partial_z)$.

Boundary conditions are periodic in the down-front direction y and no-flux in both the across-front x and vertical z directions. Forcing by a surface buoyancy loss B is implemented as a volumetric forcing within the upper-most horizontal plane and is related to the surface cooling flux density $Q = 100 \text{ Wm}^{-2}$ via nominal values of the volume expansion coefficient α_V , isobaric specific heat capacity c_p and density ρ of water and gravitational acceleration g ,

$$B = \begin{cases} -\Delta z \frac{\alpha_V g}{c_p \rho} Q & \text{if } -\Delta z < z < 0, \\ 0 & \text{otherwise,} \end{cases}$$

where Δz is the vertical grid resolution.

The full domain is comprised of two regions distinguished by their across-front resolution. An inner, high-resolution region contains the center of the front. To accommodate the large but finite initial front, the outer region has a geometrically increasing across-front grid size. The simulation state in the outer region is not discussed in this text, and results are only presented for the central region. Grid spacing is uniform in the vertical and down-front directions. The down-front resolution is such that grid spacing is horizontally isotropic within the central region. The size of the full grid is $2048 \times 128 \times 256$, with the central region comprising a sub-domain of size $1664 \times 128 \times 170$ and physical extent $5 \text{ km} \times 384 \text{ m} \times 100 \text{ m}$, corresponding to resolutions of approximately $3.0 \text{ m} \times 3.0 \text{ m} \times 0.60 \text{ m}$. Note the geometry is chosen to satisfy the thin-slice approximation. To test its validity, we ran simulations with half the across-front and vertical resolution with a variable down-front domain length. While the largest domain sizes (up to 3 km) undergo baroclinic instability during the initialization phase, this is suppressed once the background strain turns on and all simulations exhibit similar evolution.

The lower 50 m of the vertical domain contains the deep water, which is initially quiescent and with a constant stratification. This region contains a sponge layer that dampens internal waves propagating toward the bottom boundary. Specifications of the simulation grid and sponge layer are included in Texts S3 and S4 in Supporting Information S1 respectively.

The simulation state is pre-initialized with a front in thermal wind balance, with extremal Rossby and Richardson numbers $\max(\text{Ro}) = 0.4$ and $\min(\text{Ri}) = 2.0$ following the procedure described in Text S2 in Supporting Information S1. For the first 100 hr, the simulation is forced by surface buoyancy loss alone to allow the front to adjust to the mixing and reach a turbulent thermal wind balance (Gula et al., 2014; McWilliams et al., 2015). Then, we progressively increase the spatially constant strain of the background velocity field $\mathbf{U}$ following $\alpha_{\text{ramp}}(t) = \alpha \max(1 - e^{-t/15}, 0)$, with $\alpha = f/10$. We choose a modest value of α to avoid the background flow directly influencing the dynamics of the front (Shcherbina et al., 2013; Thomas, 2012). We only wish to use it to unambiguously, and somewhat naturally, create a submesoscale front. Likewise, we increase the strain slowly, over many inertial periods, in order to prevent inertia-gravity waves being generated from a sudden imbalance.

We integrate the equation set (Equations 1a–1c) using Oceananigans, a Julia package for finite-volume simulations of the non-hydrostatic Boussinesq equations in an oceanic context (Ramadhan et al., 2020; Wagner et al., 2025). Model fields are advanced with a third-order Runge-Kutta timestepper. Momentum and tracer advection is implemented using a ninth-order weighted essentially non-oscillatory (WENO) advection scheme (Durran, 2010; Shu, 1999; Silvestri et al., 2024).

2.2. Mean State Evolution

We track the evolution of the domain-averaged turbulent kinetic energy $\text{TKE} = \langle \mathbf{u}' \cdot \mathbf{u}' \rangle / 2$, where primes denote deviations from the down-front mean (i.e., average along the y -direction), and $\langle \cdot \rangle$ denotes integration over the entire domain of the simulation, although contributions from outside the central region are negligible. The TKE exchanges energy with the front and with turbulent dissipation following

$$\frac{d}{dt} \text{TKE} = \underbrace{-\langle (\partial_x \bar{\mathbf{u}}) \cdot \mathbf{u}' \mathbf{u}' \rangle}_{\text{LSP}} - \underbrace{\langle (\partial_z \bar{\mathbf{u}}) \cdot \mathbf{w}' \mathbf{u}' \rangle}_{\text{VSP}} + \underbrace{\alpha \langle u'^2 - v'^2 \rangle}_{\text{DSP'}} + \underbrace{\langle \mathbf{w}' \mathbf{b}' \rangle}_{\text{BFLUX}} + \epsilon, \quad (3)$$

where the overbars denote down-front averages (e.g., $\bar{\mathbf{u}} = \mathbf{u} - \mathbf{u}'$), and where LSP, VSP, and DSP' stand for lateral, vertical and perturbation deformation shear production, respectively (Thomas, 2012). We also introduce the integrated geostrophic shear production $\text{GSP} = \langle (\partial_x \bar{b}/f) \mathbf{w}' \mathbf{v}' \rangle$, namely, the vertical shear production that acts only on the geostrophic component of the flow, here approximated as the flow in thermal wind balance with the mean buoyancy gradient.

We calculate the turbulent flux terms $\mathbf{u}' \mathbf{u}'$, $\mathbf{w}' \mathbf{u}'$ and $\mathbf{w}' \mathbf{b}'$ using the WENO advection operators used for simulation time-stepping (Text S5 in Supporting Information S1). This implicit advection scheme produces an effective dissipation of kinetic energy ϵ , which we calculate as the residual term required for Equation 3 to hold.

The Ertel potential vorticity associated with the down-front averaged flow (hereafter simply referred to as PV), q , is useful when analyzing the stability of the flow to perturbations. A mean state with positive PV is stable to all inertia-gravity wave normal modes. A mean state with negative PV is unstable to gravitational, symmetric, inertial and Kelvin-Helmholtz instabilities (Hoskins & Bretherton, 1972; Stone, 1966; Thomas et al., 2013). For a Boussinesq fluid,

$$q = (\nabla \times \bar{\mathbf{u}} + f \hat{\mathbf{z}}) \cdot \nabla \bar{b}. \quad (4)$$

We also compute the local Richardson (Ri) number of the mean state,

$$\text{Ri} = N^2 / S^2 \quad \text{where} \quad N^2 = \partial_z \bar{b} \quad \text{and} \quad S^2 = (\partial_z \bar{\mathbf{u}})^2 + (\partial_z \bar{\mathbf{v}})^2. \quad (5)$$

3. Evolution and Processes

3.1. Overview

Figures 1 and 2 illustrate the evolution of the front over the course of the frontal steepening and arrest. The former shows snapshots of across-front velocity and buoyancy, and the latter shows Hovmöller diagrams of down-front mean across front velocity $\bar{u} + U$ at the $z = -50$ m level and near the surface. As the background forcing is applied, the front undergoes deformation-driven frontogenesis.

Frontogenesis is arrested by $t \approx 100$ hr at a width of approximately 300 m at the surface (see Figure 2b, along $x \approx -2$ km). Near the surface, there is a region hosting vigorous three-dimensional turbulence highlighted in Figures 1g and 1h, which we will refer to as the “arrest volume,” based on its apparent role in preventing further sharpening of the front, and shared position with the singularity observed in non-linear frontogenesis models of for example Hoskins and Bretherton (1972) or Shakespeare (2016). At this point the large-scale structure of the front is quasi-steady, with further motion of the isopycnals due to cooling only.

During the transient evolution, down-front symmetric tilted circulation cells grow as we turn on the deformation field and peak in intensity around $t \approx 50$ hr. Figures 1d, 1f, and 2a show that the cells predominantly align with isopycnals as they evolve, and by $t = 100$ hr (Figures 1g and 1i), they merge into a single-cell circulation that surrounds the remnants of the front below the arrest volume.

3.2. Turbulent Kinetic Energy

A comparison of the total contribution of each of the terms in the TKE balance (Equation 3), expressed relative to the total kinetic energy at $t = 0$, in both the whole domain and the arrest volume is shown in Figure 3. The

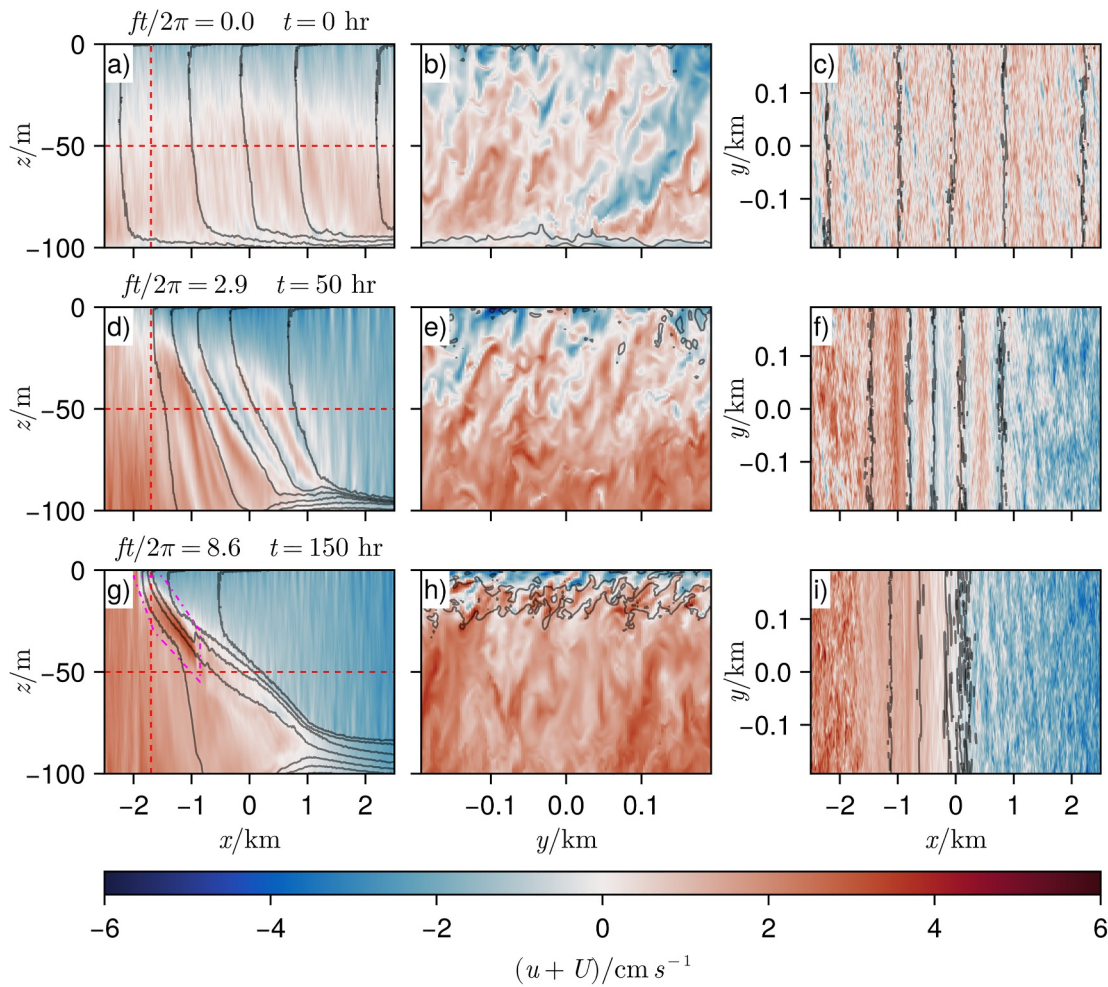


Figure 1. Evolution of the total across-front velocity (a, d, g) in the mean state $\bar{u} + U$ and slices in the (b, e, h) $y - z$ plane and (c, f, i) $x - y$ plane at positions indicated by the red, dashed lines in (a, c, e). Snapshots are shown at (a, b, c) $t = 0$ hr, (d, e, f) $t = 50$ hr and (g, h, i) $t = 150$ hr. Black solid contours show isosurfaces of (a, d, g) $\bar{b}$ and b in others. A magenta, dash-dotted contour in (g) indicates the “arrest volume” referenced in the text.

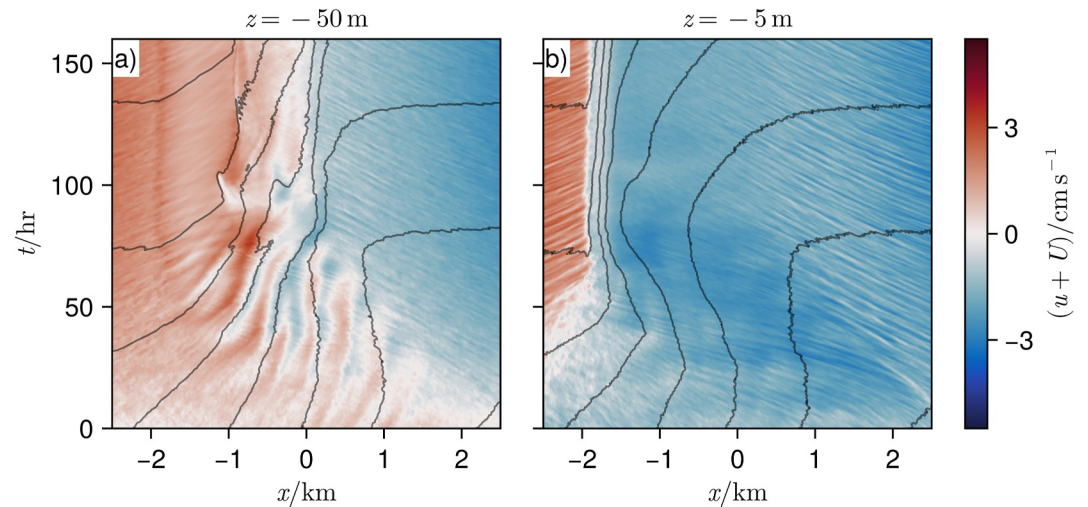


Figure 2. Hovmöller plots of the total across-front velocity in the mean state, $\bar{u} + U$, (a) at the center of the front, $z = -50$ m and (b) near the surface, $z = -5$ m. Black contours show isosurfaces of $\bar{b}(x, z, t)$.

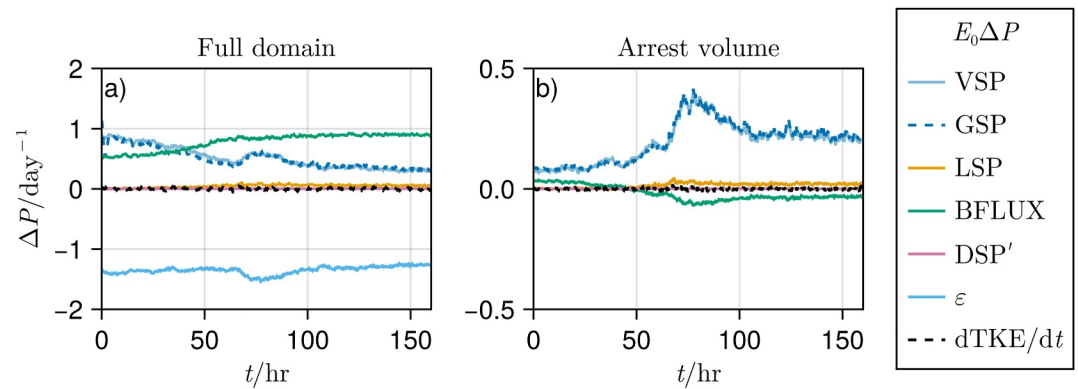


Figure 3. (a) TKE production terms in the turbulent kinetic energy balance Equation 3 and (b) the contributions to each term from the region containing the arrested front only (see Figure 1e). The terms are expressed relative to E_0 , the total kinetic energy in the front at $t = 0$.

dominant terms differ throughout the domain. VSP and BFLUX are the dominant TKE production terms, which would be expected in the submesoscale mixed layer (Brannigan, 2016; Thomas et al., 2013). The dominant mixing is due to convection and the background deformation flow is weak, with little direct effect on the energetics. When considering the whole domain (Figure 3a), we see convection increasing in importance. Indeed, VSP is roughly double BFLUX at early times, but in the final arrested state the total BFLUX is larger, while the VSP is confined to the smaller arrested volume (Figure 3b). Throughout the simulation, VSP and GSP are virtually indistinguishable and peak at $t \approx 75$ hr, coinciding with the coalescence of the down-front symmetric cells in Figure 2a.

3.3. Instability Dynamics During the Transient Phase

SI is active during the transient phase, as hinted by the dominance of VSP and GSP peaking domain-wide at $t \approx 75$ hr. Negative mean-state PV is distributed throughout the front at $t = 0$ (Figure 4a) and becomes more concentrated in the central region by $t = 50$ hr (Figure 4b). The latter coincides in time and space with the cells described earlier in this section and visible in Figures 1d, 1f, and 2a, and which are downfront-symmetric. Furthermore, $N^2 > 0$ and LSP is negligible, allowing us to attribute these cells to SI (Thomas et al., 2013). The cells redistribute negative potential vorticity to smaller horizontal scales, between the $q \approx 0$ regions in Figure 4b, where it is eventually mixed by turbulent fluxes and implicit diffusion due to the WENO scheme. This increase in potential vorticity is accompanied by isopycnal flattening below $z = -20$ m.

3.4. Arrested State

By $t = 150$ hr, most of the fluid in the front is stabilized and the primary arresting process at the surface is shear instability. Indeed, the arrest volume has strongly negative PV (Figure 4c), TKE production is mainly due to VSP or GSP, with LSP remaining negligible (Figure 3b), but motion is strongly downfront-asymmetric, with structures somewhat reminiscent of Kelvin-Helmholtz billows (Figure 1b), thereby ruling out SI. Furthermore, the Richardson number in the arrest volume is around, or much less than, 0.25 (Figure 4d and Movie S1).

Negative PV is originally created by surface cooling, advected toward the central region by the deformation flow, and concentrated near the surface by the ageostrophic secondary circulation. In the arrested state, PV integrated across the arrest volume must remain approximately constant, made possible by a constant supply of positive PV from the pycnocline, through a narrow “PV conduit” running alongside the frontal isopycnals (compare Figures 1g and 4c). Along this conduit, negative mixed-layer PV and positive pycnocline PV mix, PV monotonously decreasing from the pycnocline to the surface.

The underside of the front is only weakly stable, unlike the fluid with pycnocline PV, which saturates the color scale in Figure 4c.

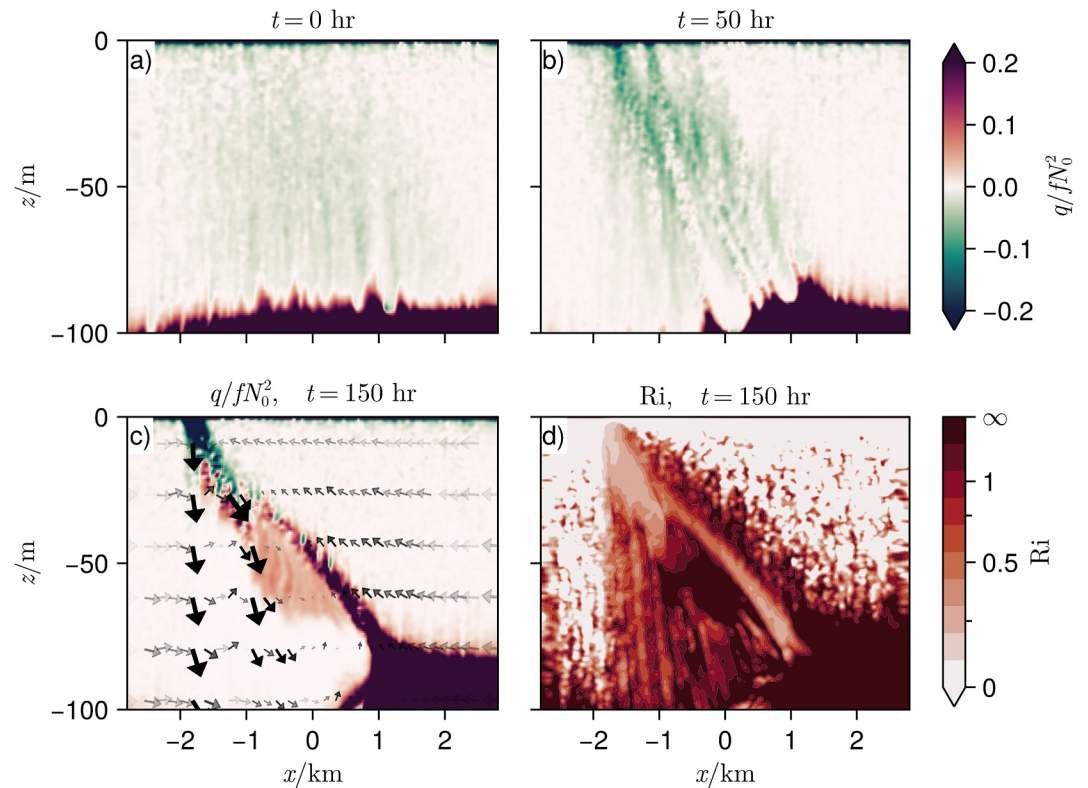


Figure 4. Potential vorticity (a, b, c) and Richardson number (d) of the down-front mean state at (a) $t = 0$, (b) $t = 50$ hr and (c, d) $t = 150$ hr. The mean secondary circulation is shown as black arrows in (c). Opacity is proportional to vertical velocity, with totally opaque arrows corresponding to a vertical velocity in excess of 80 m day^{-1} . The fields are coarse-grained with a Gaussian kernel with standard deviation 30 m in the x -direction, the secondary circulation is additionally averaged over the final 30 hr of simulation time.

4. Discussion

Based on the observations of the previous section, we can piece together a possible timeline for the evolution of the front, starting from a weakly stratified front we initialized in an actively convecting mixed layer. First, from the moment the background deformation flow starts ramping up, the front converges, and along with it, the negative-PV fluid created via PV extraction by surface buoyancy loss and associated vertical mixing. The negative PV concentrates in the central region of the simulation, which increases the susceptibility of fluid to SI. Once the front reaches a small enough size, restratification due to the secondary circulation and associated with the strengthening of symmetric cells competes with vertical mixing, limiting the mixed layer depth. This is reflected in the evolution of fronts that are initially completely mixed vertically and susceptible to SI and subsequent NIW generation via geostrophic adjustment (Atkinson et al., 2025; Wienkers et al., 2021). The cells are present before the background forcing turns on at $t = 0$, and they appear to be manifestations of the forced SI described by Taylor and Ferrari (2010). Throughout the transient evolution, SI keeps the flow in a nearly marginally stable state with near-zero PV (Figures 2a and 4b). Also note that in this marginally stable state, the distinction between slow-growing SI and slow-oscillating NIWs is somewhat irrelevant (Mooers, 1975; Whitt & Thomas, 2013), and both are similarly modified by background strain (Thomas, 2012). Regardless of the nature of the cells, they eventually coalesce, resulting in a single secondary circulation cell around the portion of the front below the arrest volume.

At later times, in the arrested state, the front contain a persistently energetic, finite-size arrest volume at the surface. The region is clearly separated from the remainder of the front and well-resolved. Visual inspection of the simulation timeseries (Movie S1) shows that the arrest volume is highly turbulent, hosting what appears to be shear, perhaps Kelvin-Helmholtz, instability, along with a vigorous secondary circulation. There is a sharp transition where it meets the remainder of the front, resembling a hydraulic jump around $(x, z) \approx (-1 \text{ km}, -40 \text{ m})$ in

Figure 1e, whose precise nature will require further investigation. We found, but do not show here, that these features—the location and size of the arrest region and formation of a jump—were consistent when varying the horizontal and vertical grid resolutions. The constant supply of surface-generated negative PV through the lateral and surface boundaries is compensated by a constant supply of positive pycnocline PV, funneled through an approximately along-isopycnal conduit appearing to block vertical mixing, preserving a stable region underneath the front, with intermediate stratification and weakly positive PV, subducted underneath the front by the secondary circulation (Movie S1). Note that even though the arrest volume is dominated by turbulence, its formation and maintenance is dependent on the presence of the background strain.

5. Summary

We have described an idealized large-eddy simulation of a front undergoing frontogenesis driven by a balanced background deformation flow in the presence of vertical mixing caused by surface buoyancy loss. Using a scale separation argument, the thin-slice approximation allowed for along-front periodic boundary conditions while preserving the frontogenetic effect of the deformation flow forcing and the three-dimensional character of the turbulence. We believe that these results contain the first controlled, clearly forced LES demonstration of a finite frontal arrest region as the clear result of a strain forcing applied to a front. The high resolution of our simulations is such that the front can cross a range of scales as it evolves, and we observe a qualitative change in regime from a frontal-scale secondary circulation throughout the depth of the domain to a state characterized by a small—but still resolved by a few dozen grid cells in each direction—region at the position of the arrested surface front characterized by shear instability and turbulence. We also observe a conduit that entrains fluid from the base of the mixed layer, along isopycnals, toward this turbulent region. The onset of the second regime is marked by the strengthening of symmetric cells in the secondary circulation. We argue that vertical mixing in the final arrested state is limited by restratification due to both the frontogenetic secondary circulation and stabilization by entrained pycnocline PV. This positive PV acts as a barrier, preventing vertical mixing and creating a quiescent region of marginally-stable fluid below the front.

The evolution of the front, onset of a transient SI-like2 circulation and mechanisms of frontogenesis and frontal arrest within the steady, arrested front are rich avenues of future exploration to further understanding of the role that fronts play in the global ocean and we hope to address each of these areas in more detail in future work. In particular, the method used here can be extended to generate an ensemble of simulations of frontogenesis in the presence of surface forcing by cooling or winds in a controlled manner to study the arrest region width, restratification and their dependence on air-sea heat and momentum fluxes and the larger-scale ocean flow.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

Simulation and figure setup scripts are available at <https://doi.org/10.5281/zenodo.20024831> (Atkinson, 2026). Simulations were performed using Oceananigans v0.101.0 (Ramadhan et al., 2020; Wagner et al., 2025). Figures used in the paper were generated using Makie v0.24.8 (Danisch & Krumbiegel, 2021).

References

- Aravind, H. M., Verma, V., Sarkar, S., Freilich, M. A., Mahadevan, A., Haley, P. J., et al. (2023). Lagrangian surface signatures reveal upper-ocean vertical displacement conduits near oceanic density fronts. *Ocean Modelling*, 181, 102136. <https://doi.org/10.1016/j.ocemod.2022.102136>
- Atkinson, E. (2026). Formation and arrest of a surface density front via strain-driven frontogenesis - Setup and analysis [Software]. *Zenodo*. <https://doi.org/10.5281/zenodo.20024831>
- Atkinson, E., McWilliams, J. C., & Grisouard, N. (2025). Near-inertial echoes of ageostrophic instability in submesoscale filaments. *Journal of Fluid Mechanics*, 1015, A17. <https://doi.org/10.1017/jfm.2025.10348>
- Barkan, R., Molemaker, M. J., Srinivasan, K., McWilliams, J. C., & D'Asaro, E. A. (2019). The role of horizontal divergence in submesoscale frontogenesis. *Journal of Physical Oceanography*, 49(6), 1593–1618. <https://doi.org/10.1175/JPO-D-18-0162.1>
- Blumen, W. (2000). Inertial oscillations and frontogenesis in a zero potential vorticity model. *Journal of Physical Oceanography*, 30(1), 31–39. [https://doi.org/10.1175/1520-0485\(2000\)030<0031:IOAFIA>2.0.CO;2](https://doi.org/10.1175/1520-0485(2000)030<0031:IOAFIA>2.0.CO;2)
- Brannigan, L. (2016). Intense submesoscale upwelling in anticyclonic eddies. *Geophysical Research Letters*, 43(7), 3360–3369. <https://doi.org/10.1002/2016GL067926>
- Brüggemann, N., & Eden, C. (2015). Routes to dissipation under different dynamical conditions. *Journal of Physical Oceanography*, 45(8), 2149–2168. <https://doi.org/10.1175/JPO-D-14-0205.1>

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- Crowe, M. N., & Taylor, J. R. (2019). The evolution of a front in turbulent thermal wind balance. Part 2. Numerical simulations. *Journal of Fluid Mechanics*, 880, 326–352. <https://doi.org/10.1017/jfm.2019.688>
- Danisch, S., & Krumbiegel, J. (2021). Makie.jl: Flexible high-performance data visualization for Julia. *Journal of Open Source Software*, 6(65), 3349. <https://doi.org/10.21105/joss.03349>
- Dauhajre, D. P., Srinivasan, K., Molemaker, M. J., Gula, J., Hypolite, D., McWilliams, J. C., et al. (2025). Vertical mixing can both induce and inhibit submesoscale frontogenesis. *Journal of Physical Oceanography*, 55, 1409–1433. <https://doi.org/10.1175/JPO-D-24-0148.1>
- Durran, D. R. (2010). *Numerical methods for fluid dynamics: With applications to geophysics* (Vol. 32). Springer Science & Business Media. Retrieved from <http://www.springer.com/series/1214>
- Ferrari, R., & Wunsch, C. (2009). Ocean circulation kinetic energy: Reservoirs, sources, and sinks. *Annual Review of Fluid Mechanics*, 41, 253–282. <https://doi.org/10.1146/annurev.fluid.40.111406.102139>
- Freilich, M., & Mahadevan, A. (2021). Coherent pathways for subduction from the surface mixed layer at ocean fronts. *Journal of Geophysical Research: Oceans*, 126(5), e2020JC017042. <https://doi.org/10.1029/2020JC017042>
- Gula, J., Molemaker, M. J., & McWilliams, J. C. (2014). Submesoscale cold filaments in the Gulf Stream. *Journal of Physical Oceanography*, 44(10), 2617–2643. <https://doi.org/10.1175/JPO-D-14-0029.1>
- Gula, J., Taylor, J., Shcherbina, A., & Mahadevan, A. (2021). Submesoscale processes and mixing. In *Ocean mixing: Drivers, mechanisms and impacts* (pp. 181–214). Elsevier. <https://doi.org/10.1016/B978-0-12-821512-8.00015-3>
- Hoskins, B., & Bretherton, F. (1972). Atmospheric frontogenesis models: Mathematical formulation and solution. *Journal of the Atmospheric Sciences*, 29, 11–37. [https://doi.org/10.1175/1520-0469\(1972\)029<0011:afmmfa>2.0.co;2](https://doi.org/10.1175/1520-0469(1972)029<0011:afmmfa>2.0.co;2)
- Loken, C., Gruner, D., Groer, L., Peltier, R., Bunn, N., Craig, M., et al. (2010). Scinet: Lessons learned from building a power-efficient top-20 system and data centre. *Journal of Physics: Conference Series*, 256, 012026. <https://doi.org/10.1088/1742-6596/256/1/012026>
- Mahadevan, A. (2016). The impact of submesoscale physics on primary productivity of plankton. *Annual Review of Marine Science*, 8(1), 161–184. <https://doi.org/10.1146/annurev-marine-010814-015912>
- McWilliams, J. C. (2016). Submesoscale currents in the ocean. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Science*, 472(2189), 20160117. <https://doi.org/10.1098/rspa.2016.0117>
- McWilliams, J. C. (2021). Oceanic frontogenesis. *Annual Review of Marine Science*, 13(1), 227–253. <https://doi.org/10.1146/annurev-marine-032320-120725>
- McWilliams, J. C., Gula, J., Molemaker, M. J., Renault, L., & Shchepetkin, A. F. (2015). Filament frontogenesis by boundary layer turbulence. *Journal of Physical Oceanography*, 45(8), 1988–2005. <https://doi.org/10.1175/jpo-d-14-0211.1>
- Mooers, C. N. K. (1975). Several effects of a baroclinic current on the cross-stream propagation of inertial-internal waves. *Geophysical Fluid Dynamics*, 6(3), 245–275. <https://doi.org/10.1080/03091927509365797>
- Pham, H. T., Verma, V., Sarkar, S., Shcherbina, A. Y., & D'Asaro, E. A. (2024). Rapid downwelling of tracer particles across the boundary layer and into the pycnocline at submesoscale ocean fronts. *Geophysical Research Letters*, 51(17), e2024GL109674. <https://doi.org/10.1029/2024GL109674>
- Ponce, M., van Zon, R., Northrup, S., Gruner, D., Chen, J., Ertinaz, F., et al. (2019). Deploying a top-100 supercomputer for large parallel workloads. In *Proceedings of the practice and experience in advanced research computing on rise of the machines (learning)* (pp. 1–8). ACM. <https://doi.org/10.1145/3332186.3332195>
- Ramadhan, A., Wagner, G., Hill, C., Campin, J.-M., Churavy, V., Besard, T., et al. (2020). Oceananigans.jl: Fast and friendly geophysical fluid dynamics on GPUS. *Journal of Open Source Software*, 5(53), 2018. <https://doi.org/10.21105/joss.02018>
- Samelson, R. M., & Skillingstad, E. D. (2016). Frontogenesis and turbulence: A numerical simulation. *Journal of the Atmospheric Sciences*, 73(12), 5025–5040. <https://doi.org/10.1175/jas-d-16-0145.1>
- Shakespeare, C. J. (2016). Nonhydrostatic wave generation at strained fronts. *Journal of the Atmospheric Sciences*, 73(7), 2837–2850. <https://doi.org/10.1175/JAS-D-14-0294.1>
- Shcherbina, A. Y., D'Asaro, E., Lee, C. M., Klymak, J. M., Molemaker, M. J., & McWilliams, J. C. (2013). Statistics of vertical vorticity, divergence, and strain in a developed submesoscale turbulence field. *Geophysical Research Letters*, 40(17), 4706–4711. <https://doi.org/10.1002/grl.50919>
- Shu, C.-W. (1999). High order ENO and WENO schemes for computational fluid dynamics. In *High-order methods for computational physics* (pp. 439–582). Springer.
- Silvestri, S., Wagner, G. L., Campin, J. M., Constantinou, N. C., Hill, C. N., Souza, A., & Ferrari, R. (2024). A new WENO-based momentum advection scheme for simulations of ocean mesoscale turbulence. *Journal of Advances in Modeling Earth Systems*, 16, e2023MS004130. <https://doi.org/10.1029/2023MS004130>
- Srinivasan, K., Barkan, R., & McWilliams, J. C. (2023). A forward energy flux at submesoscales driven by frontogenesis. *Journal of Physical Oceanography*, 53(1), 287–305. <https://doi.org/10.1175/JPO-D-22-0001.1>
- Stone, P. H. (1966). On non-geostrophic baroclinic stability. *Journal of the Atmospheric Sciences*, 23(4), 390–400. [https://doi.org/10.1175/1520-0469\(1966\)023<0390:ONGBS>2.0.CO;2](https://doi.org/10.1175/1520-0469(1966)023<0390:ONGBS>2.0.CO;2)
- Sullivan, P. P., & McWilliams, J. C. (2024). Oceanic frontal turbulence. *Journal of Physical Oceanography*, 54, 333–358. <https://doi.org/10.1175/JPO-D-23-0033.1>
- Taylor, J. R., & Ferrari, R. (2010). Buoyancy and wind-driven convection at mixed layer density fronts. *Journal of Physical Oceanography*, 40, 1222–1242. <https://doi.org/10.1175/2010JPO4365.1>
- Taylor, J. R., & Thompson, A. F. (2022). Submesoscale dynamics in the upper ocean. *Annual Review of Fluid Mechanics*, 55, 103–127. <https://doi.org/10.1146/annurev-fluid-031422>
- Thomas, L. N. (2005). Destruction of potential vorticity by winds. *Journal of Physical Oceanography*, 35(12), 2457–2466. <https://doi.org/10.1175/JPO2830.1>
- Thomas, L. N. (2012). On the effects of frontogenetic strain on symmetric instability and inertia-gravity waves. *Journal of Fluid Mechanics*, 711, 620–640. <https://doi.org/10.1017/jfm.2012.416>
- Thomas, L. N., Taylor, J. R., Ferrari, R., & Joyce, T. M. (2013). Symmetric instability in the Gulf Stream. *Deep-Sea Research Part II Topical Studies in Oceanography*, 91, 96–110. <https://doi.org/10.1016/j.dsr2.2013.02.025>
- Wagner, G. L., Silvestri, S., Constantinou, N. C., Ramadhan, A., Campin, J.-M., Hill, C., et al. (2025). High-level, high-resolution ocean modeling at all scales with oceananigans. (submitted to the Journal of Advances in Modeling Earth Systems). Retrieved from <http://arxiv.org/abs/2502.14148>
- Whitt, D. B., & Thomas, L. N. (2013). Near-inertial waves in strongly baroclinic currents. *Journal of Physical Oceanography*, 43(4), 706–725. <https://doi.org/10.1175/JPO-D-12-0132.1>
- Wieners, A. F., Thomas, L. N., & Taylor, J. R. (2021). The influence of front strength on the development and equilibration of symmetric instability. Part 1. Growth and saturation. *Journal of Fluid Mechanics*, 926, A6. <https://doi.org/10.1017/jfm.2021.680>